

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location Amazon Web Services, OH -- station KCMH, America/New_York
Committed pair OSM 1386016629 -> 1386016631
Amazon Web Services / Amazon Web Services
Facade gap 73.3 m between the two halls
Weather record 43,786 real hourly records from KCMH
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3503 C at 110 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2024-02-28 (crossing)
Plant limit 18.0 C
Notice required 6 h
Forecast skill 0.00 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 8 of 24 hours with 1 mode change. That is 3 more than the reactive incumbent, at 0 unsafe hours against its 2. 5 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 8 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 5 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s)
5 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dew point	15.685	18.0	16.110	3.933
01	mechanical	no	dew point	15.757	18.0	16.110	3.510
02	mechanical	no	dew point	16.557	18.0	16.670	3.542
03	mechanical	yes	switch budget	16.168	18.0	18.330	3.065
04	mechanical	yes	switch budget	16.557	18.0	18.890	2.769
05	mechanical	yes	switch budget	16.209	18.0	17.220	2.401
06	mechanical	yes	switch budget	16.722	18.0	15.000	2.449

07	mechanical	yes	switch budget	17.263	18.0	15.000	2.073
08	mechanical	no	dry-bulb	20.093	18.0	14.440	2.691
09	mechanical	no	dry-bulb	24.479	18.0	10.560	3.655
10	mechanical	no	dry-bulb	27.792	18.0	6.670	4.726
11	mechanical	no	dry-bulb	28.363	18.0	4.440	5.596
12	mechanical	no	dry-bulb	27.247	18.0	3.330	5.766
13	mechanical	no	dry-bulb	27.273	18.0	2.780	5.623
14	mechanical	no	dry-bulb	25.058	18.0	1.670	4.735
15	mechanical	no	dry-bulb	19.523	18.0	1.670	4.359
16	FREE	yes	-	12.822	18.0	0.560	3.134
17	FREE	yes	-	8.920	18.0	0.000	3.136
18	FREE	yes	-	5.603	18.0	-1.110	2.536
19	FREE	yes	-	3.922	18.0	-1.670	2.912
20	FREE	yes	-	1.760	18.0	-2.220	3.270
21	FREE	yes	-	1.166	18.0	-2.780	3.583
22	FREE	yes	-	0.044	18.0	-3.890	4.017
23	FREE	yes	-	-0.526	18.0	-3.890	4.061

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.685 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

01:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.757 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

02:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.557 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.167 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.557 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.209 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is

also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.722 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.263 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.093 C against a 18.0 C limit, so it fails by 2.093 C. It would take a limit 2.093 C higher, or a bound 2.093 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.479 C against a 18.0 C limit, so it fails by 6.479 C. It would take a limit 6.479 C higher, or a bound 6.479 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.792 C against a 18.0 C limit, so it fails by 9.792 C. It would take a limit 9.792 C higher, or a bound 9.792 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.363 C against a 18.0 C limit, so it fails by 10.363 C. It would take a limit 10.363 C higher, or a bound 10.363 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.247 C against a 18.0 C limit, so it fails by 9.247 C. It would take a limit 9.247 C higher, or a bound 9.247 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.273 C against a 18.0 C limit, so it fails by 9.273 C. It would take a limit 9.273 C higher, or a bound 9.273 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.058 C against a 18.0 C limit, so it fails by 7.058 C. It would take a limit 7.058 C higher, or a bound 7.058 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.523 C against a 18.0 C limit, so it fails by 1.523 C. It would take a limit 1.523 C higher, or a bound 1.523 C tighter, to change this hour.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.822 C, which is 5.178 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.134 C, and the margin is measured, not chosen: 3.102 C of group-conditional forecast error for this hour of day, plus 0.0319 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.920 C, which is 9.080 C

under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.136 C, and the margin is measured, not chosen: 3.096 C of group-conditional forecast error for this hour of day, plus 0.0398 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.603 C, which is 12.397 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.536 C, and the margin is measured, not chosen: 2.494 C of group-conditional forecast error for this hour of day, plus 0.0425 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 3.922 C, which is 14.078 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.912 C, and the margin is measured, not chosen: 2.880 C of group-conditional forecast error for this hour of day, plus 0.0322 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 1.760 C, which is 16.240 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.270 C, and the margin is measured, not chosen: 3.181 C of group-conditional forecast error for this hour of day, plus 0.0898 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 1.166 C, which is 16.834 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.583 C, and the margin is measured, not chosen: 3.527 C of group-conditional forecast error for this hour of day, plus 0.0563 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.044 C, which is 17.956 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.017 C, and the margin is measured, not chosen: 3.973 C of group-conditional forecast error for this hour of day, plus 0.0445 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.526 C, which is 18.526 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.061 C, and the margin is measured, not chosen: 4.027 C of group-conditional forecast error for this hour of day, plus 0.0336 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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